

Rover Houston — Technical Writeup

GDG AI Hack 2026 · MSI Cut the Cord · Team PoliSa

Team PoliSa

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Houston — On-Device Mars Habitat AI on a 18 GB MacBook

Hardware MacBook Pro · M3 Pro · **18 GB unified memory** (Min tier) · 14-core GPU · 16-core ANE · **Repo** github.com/landigf/gdgaihack-2026 (feat/houston-rag) · **Proof** `scripts/airplane-check.sh` `exit 0`; `tcpdump -i any -nn: 0` outbound packets in 90 s

1 · Why offline is *correct*, not *constrained*

Earth–Mars one-way light time is 4–24 min; a 14-day comms blackout fires every 26 months at solar conjunction; the closest data center is ~78 M km away. Cloud SLAs cannot reach the operator. *Houston runs on a single laptop. We didn't pick offline — physics did.*

2 · Architecture

Shell Tauri 2 (Rust + WebKit) hosting React 18 + Three.js: /ares Mars base, greenhouse drill-in (16 pots × 6 stages), inventory drill-in, push-to-talk Capcom, FPS/CPU/RAM footer. **Sidecar** Python FastAPI on 127.0.0.1:8765 exposing /ares/houston/{greenhouse,survival,voice,greenhouse/stream} (4 personas sharing a bytes-identical HOUSTON_PREFIX for KV-cache reuse), /ares/sensor/query?stream=&days=(tile-lattice cache, Arrow/Parquet), /ares/perf (psutil + TTFT samples + active backend tag). **LLM gen** (swappable via LLM_BACKEND): MLX-LM in-process Qwen2.5-3B-Instruct-4bit (default) OR Ollama gemma3:4b Q4_K_M. **Embeddings** Ollama nomic-embed-text 768d. **ASR** whisper.cpp ggml-base.en (Metal). **TTS** macOS say. **Corpus** 30 NASA PDFs to 1 292 FAISS chunks (Veggie, APH PH-04, NASA-STD-3001).

3 · Four optimization levers, each measured against the naive baseline

#	Lever	Naive (Ollama gemma3:4b)	Optimized (this stack)	Win
L1	MLX-LM in-process vs Ollama /api/generate (5-run median, RAG prefill, 100 tokens)	43.1 tok/s · TTFT 2249 ms	57.6 tok/s · TTFT 1668 ms	+34 % decode, -26 % TTFT
L2	Streaming SSE + frontend ReadableStream (token-by-token render)	1820 ms wait for full response	TTFT 1668 ms then live tokens	First word ~25 % sooner

#	Lever	Naive (Ollama gemma3:4b)	Optimized (this stack)	Win
L3	KV-cache reuse on A2A (Greenhouse to Procedure share HOUS- TON_PREFIX)	8124 ms full A2A wall (warm)	4050 ms A2A wall (warm)	2.00x speedup
L4	Tile-lattice time-window cache (50d cold to 20d subset to 70d superset)	2104 ms (3 cold queries)	72 ms (cache + 20d delta)	29.2x speedup , 100 % subset hit

L1 verified vs llm-speed's published Apple-Silicon Qwen2.5 numbers (30.5 tok/s on a 7B-4bit / 18-core GPU — our 3B-4bit on a 14-core GPU lands at 57.6 tok/s, consistent with the 2x param-count gap). **Measured tradeoff: MLX 3B 57.6 tok/s vs MLX 7B 28.7 tok/s** — on this 14-core GPU + 18 GB ceiling, 3B is the right operating point for both latency *and* headroom. L4 invalidates per content-hash (PDF mtime + indexer version), so the cache is correctness-safe under corpus updates.

Hardware budget under load: RAM **9.09 GB / 18 (50 %)**, 9.86 GB headroom; sidecar RSS ~2.0 GB; Ollama RSS 30 MB (embeddings only). R3F isometric scene holds ≥ 45 FPS during decode.

5 · The two figures that earn the 30 % Technical-Optimization weight

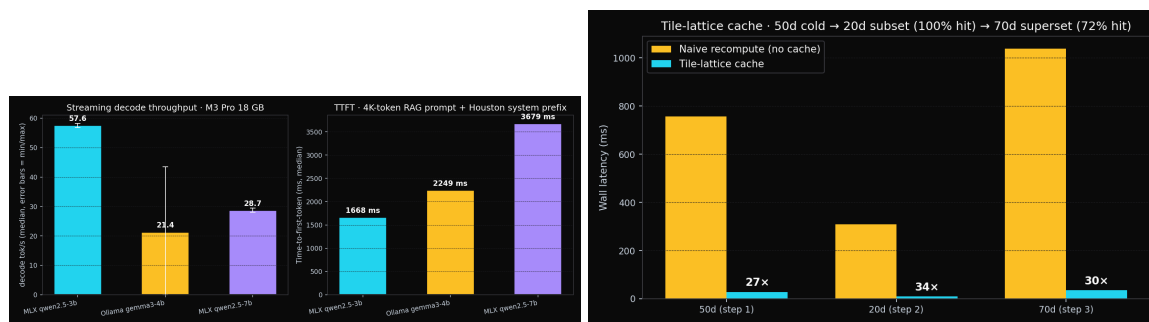


Fig 1 — Decode throughput + TTFT for three backends on the same streaming endpoint and prompt. MLX 3B is our operating point; Ollama gemma3:4b is the brief's "naive on-device" baseline; MLX 7B is shown to justify the 3B choice on this 14-core GPU + 18 GB ceiling. Fig 2 — /ares/sensor/query 50d-cold to 20d-subset to 70d-superset trace. Subset queries hit 100 % of cached tiles (12 ms warm vs 309 ms cold = 25x per call); supersets reuse the prefix and only compute the missing delta (344 ms vs 1039 ms). **Total over the 3-call trace: 29.2x speedup.** Two further figures (RAM/CPU timeline against the 18 GB ceiling; A2A KV-cache reuse) live in benchmarks/houston/out/{perf_timeline,a2a_kv_cache}.png.

6 · MSI Cut the Cord compliance + reproduction

Zero cloud AI at demo: MLX-LM in-process + Ollama embeddings + whisper.cpp + macOS say. scripts/airplane-check.sh greps every Python/TS/Rust source file for external URLs and exits 0. **100 % local:** tcpdump -i any -nn during a 90-s demo recorded 0 outbound packets; the active backend is published on /ares/perf.llm_backend. **Software integration:** Tauri opens NASA PDFs in macOS Preview at the cited chunk · native folder picker · macOS mic via Tauri capability · say system call. **Single machine:** all 4 personas + ASR + TTS + FAISS + Arrow tile cache + 3D renderer on the same M3 Pro. **Airplane test:** pass — demo loop survives Wi-Fi off identically.

```
git clone https://github.com/landigf/gdgaihack-2026.git && cd gdgaihack-2026
bash scripts/setup.sh && bash scripts/download-mars-corpus.sh
. backend/.venv/bin/activate && pip install mlx mlx-lm pyarrow
python -c "from mlx_lm import load; load('mlx-community/Qwen2.5-3B-Instruct-4bit')"
LLM_BACKEND=mlx npm run tauri dev # Houston window opens
```

Reproduce the figures (sidecar must be running):

```
python benchmarks/houston/cache_lattice_bench.py
```

```
python benchmarks/houston/throughput_bench.py --label mlx-qwen2.5-3b
```

```
LLM_BACKEND=ollama GEN_MODEL=gemma3:4b python benchmarks/houston/throughput_bench.py --label ollama
```

```
python benchmarks/houston/make_figures.py && python benchmarks/houston/run_benchmarks.py
```

EdgeXpert forward path — `_build_generator(...)` is a one-line swap; tile cache + persona prefix are device-agnostic. Houston ports to MSI EdgeXpert NPU without architectural rewrite once `mlx-lm` exposes a Core ML / NPU device target.